

“KENDRIYA VIDYALAYA SANGATHAN, Varanasi Region”

Pre-Board - I Examination 2024-25

Class : X

Subject : Mathematics

Max. Marks : 80

Max. Time : 3 Hours

MARKING SCHEME

	SECTION - A (20 X 1 = 20)	
1	(b) $(5 \times 2)^n$	1
2	(a) $x^2 + 5x + 6$	1
3	(d) $a_1/a_2 = b_1/b_2 \neq c_1/c_2$	1
4	(d) -14	1
5	(a) 32	1
6	(a) 45	1
7	(b) 5	1
8	(b) $(-2, -3)$	1
9	(b) $6/7$	1
10	(b) 50°	1
11	(a) $3/5$	1
12	(b) $12/25$	1
13	(a) 60^0	1
14	(c) 4 times	1
15	(d) 216 cm^3	1
16	(b) mode = 3 median – 2 mean	1
17	(a) $1/7$	1
18	(a) $1/2$	1

19	(b) Both assertion (A) and reason (R) are true and reason (R) is not the correct explanation of assertion (A)	1
20	(a) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A)	1
	SECTION - B(5 X 2 = 10)	
21	$84 = 2^2 \times 3 \times 7$ $144 = 2^4 \times 3^2$ Common factors: $2^2 \times 3 = 12$ HCF(84, 144) = 12	1/2 1/2 1/2 1/2
22	$AD / DB = AE / EC$ $1.5 / 4.5 = 2 / EC$ $EC = 6 \text{ cm}$ OR Yes, ΔABC and ΔPQR are similar triangles. Since corresponding angles are equal, the triangles are similar by AA similarity criterion.	1/2 1/2 1 or 1 1
23	Angle swept in 5 minutes = $(360^\circ / 60) \times 5 = 30^\circ$ Area swept (A) = $(\theta / 360) \times \pi r^2$ = $(30 / 360) \times (22/7) \times 7^2$ = 12.83 cm^2 OR Angle swept in 5 minutes = $(360^\circ / 60) \times 5 = 30^\circ$ Arc length (L) = $(\theta/360) \times 2\pi r$ = $(30/360) \times 2 \times (22/7) \times 7$ = 3.67 cm	1/2 1/2 1/2 1/2 or 1/2 1/2 1/2 1/2
24	Correct Fig Correct Proof	1/2 3/2
25	$\tan A = 1$ $2\tan^2 A + 1 = 2(1)^2 + 1$ $= 2(1) + 1$ $= 2 + 1 = 3$	1/2 1/2 1/2 1/2
	SECTION - C(6 X 3 = 18)	
26	Correct Proof	3
27	Zeroes of the polynomial: $\alpha = 2, \beta = 3$ Verification of relationship between zeroes and coefficients	2 1
28	For finding value of $x = 1, y = 2$	3
29	Given ,To prove , Figure	1

	Correct Proof	2
30	Using correct identity Correct proof Or Using correct identity Correct proof	1 2 or 1 2
31	For correct table For correct formula and mean = 24.2	1 2
	SECTION - D(4 X 5 = 20)	
32	Let usual speed = x km/h $(1500 / x) - 0.5 = 1500 / (x + 100)$ $x^2 + 100x - 300000 = 0$ $x = 500$ or $x = -600$ Plane's usual speed is 500 km/h. OR Let the original speed be x km/h. $(360 / x) - (360 / (x + 5)) = 48/60$ $x^2 + 5x - 2250 = 0$ $x = 45$ or $x = -50$ Therefore original speed is 45 km/h	2 1 1 1 or 2 1 1 1
33	For correct statement Given ,To prove , Figure For correct proof	1 1 3
34	Volume of hemisphere = $(2/3)\pi r^3 = (2/3) \times 3.14 \times 2^3 = 16.755 \text{ cm}^3$ Volume of cone = $(1/3)\pi r^2 h = (1/3) \times 3.14 \times 2^2 \times 2 = 8.377 \text{ cm}^3$ Total Volume of Toy = 25.132 cm^3 Volume of cylinder = $\pi r^2 h = 3.14 \times 2^2 \times 4 = 50.24 \text{ cm}^3$ OR Height (h) = 6 × Radius (r) Total surface area (TSA) of cylinder = $2\pi r(h + r)$ $= 2 \times 22/7 \times r(6r + r) = 2 \times 22/7 \times 7r^2$ Cost = Rate × Area $237.60 = 0.60 \times 2 \times 22r^2$ $r^2 = 9$ $r = 3$ Height (h) = 6r = $6 \times 3 = 18$	3/2 3/2 1/2 3/2 or 1/2 1/2 1 1 1 1 1 1
35	For correct table $x + y = 38$ 30-40 is the median class For correct formula	1 1 1/2 1/2

	$32=30+(100-70-x)/6$ $x=18, y=20$	1 1
	SECTION - E(3 X 4 = 12)	
36	(i) Rs 4900 (ii) 10:49 (iii) Rs 73,500 OR 39,000	1 1 2
37	(i) 90 m (ii) $60\sqrt{3}$ m (iii) $100 / (1 + \sqrt{3})$ m OR $50 / (1 + \sqrt{3})$ m	1 1 2
38	(i) $\sqrt{17}$ (ii) $\sqrt{20}$ (iii) $\sqrt{29}$, Scalene triangle OR $\sqrt{17} + \sqrt{20} + \sqrt{29}$	1 1 2